8-Week Project Proposal: EarthTeam Stars Verification Dashboard MVP

UChicago Tech Team — Next Quarter (8 weeks)

1. Overview

EarthTeam Stars is a blockchain-based utility/impact token intended to track, verify, and reward collaboration, action, and impact in planetary health. Our team proposes delivering an internal Minimum Viable Product (MVP) that supports (1) submitting Action/Impact "Report Cards", (2) verifying them via a Verifier Dashboard, and (3) issuing Stars on Solana devnet as a proof-of-flow.

2. MVP Scope (What we will build)

- Reporter Portal: submit an Action (Silver) or Impact (Gold) report card with "show & tell" evidence (photos/files), outputs/outcomes, and witnesses.
- Verifier Dashboard: queue of submissions, rubric-based scoring, approve/reject, comments, and a simple reputation/track-record view.
- Scoring Engine: configurable Star ranges and thresholds (so leadership can adjust rules without code changes).
- On-chain Proof (Devnet): after approval, the system issues/ transfers an SPL token on Solana devnet and stores a memo/hash that links back to the report card record.
- Exports: CSV/JSON export of verified actions for reporting, proposals, and audits.

3. Out of Scope (Realistic limits for an 8-week student team)

- Production deployment, contract audits, and mainnet token launches.
- Full DAO governance, staking/slashing economics, or KYC/AML workflows.
- AI "Oracle" automation for media authenticity/ecological change detection (we can leave placeholders).

4. Milestones & Weekly Plan

Week	Focus	Key Tasks	Deliverable / Demo
1	Onboarding + Requirements	Read EarthTeam Stars docs; align on MVP; define user stories (Reporter, Verifier, Admin);	1–2 page Product Spec + Architecture sketch; backlog prioritized.

		choose stack; set up repo/CI; create data model draft.	
2	UX + Project Setup	Wireframes; implement login (basic); scaffold Reporter Portal; decide evidence storage; create seed data using provided scenarios.	Clickable prototype + running web app skeleton.
3	Action Report Cards (Silver) End-to-End (Off-chain)	Build Action submission form; evidence upload; witness fields; admin view; basic validation; unit tests for scoring inputs.	Demo: submit Action report card → appears in verifier queue.
4	Verifier Dashboard + Scoring Engine	Implement verifier queue, detail view, rubric scoring, approve/reject; compute Star award; audit log; configurable thresholds (# verifications).	Demo: submit → verify (5 verifiers) → final Star amount computed.
5	Solana Devnet Integration (Proof of Issuance)	Create/attach devnet SPL token mint; implement mint/transfer service; write transaction memo linking to report card record/hash; display explorer link.	Demo: approval triggers devnet transaction; show tx id + linked report card.

6	Impact (Gold) Extension + Templates	Add Impact report form fields (before/after, datasets); add scoring template per intervention category; refine verifier rubric for impact.	Demo: impact report card workflow + sample scenario verification.
7	Hardening + User Testing	Bug fixes; permissions review; improve UX; add exports (CSV/JSON); run a pilot with 2–3 EarthTeam reviewers and iterate.	Release Candidate + short user guide.
8	Final Delivery + Handoff	Final demo; documentation (setup, admin, verifier); backlog of "next quarter" improvements; handoff meeting with EarthTeam.	Deployed MVP + documentation + recorded walkthrough.

5. Team Structure (suggested)

- Project Lead / PM (1): liaison with EarthTeam, weekly planning, acceptance criteria.
- Frontend (2): Reporter Portal + Verifier UI (React/Next).
- Backend (2): API, database, auth, storage, scoring engine, exports.
- Blockchain Integration (1): Solana devnet SPL token issuance, memo/hash linking.
- QA/Docs (1): testing, demo scripts, documentation, onboarding guides.

6. What we need from EarthTeam (lightweight)

- One point of contact for weekly 30-min check-ins.
- A decision on the MVP scoring rules to implement first (we can make ranges configurable).
- A small set of sample "Action" submissions and 2–3 "Impact" examples (or permission to use anonymized samples).

- A devnet wallet + token mint authority (or approval for us to create a devnet mint for the demo).
- Feedback from 2–3 intended verifiers to test usability in Week 7.

7. Success Criteria

- A user can submit an Action report card with evidence and witnesses.
- At least 5 verifiers can review, score, and approve/reject a submission.
- The system computes a Star award and records an audit trail.
- The system issues a devnet transaction (SPL token transfer/mint) and links it to the report card via memo/hash.
- EarthTeam can export verified actions for reporting.

8. Risks & Mitigations

Scope creep: Keep MVP narrow; weekly demos; strict "out of scope" list.

Unclear/ evolving scoring rules: Make rules configurable; document assumptions; request weekly sign-off on rule changes.

Blockchain complexity for beginners: Use devnet + existing SPL primitives; avoid custom Solana programs.

Security concerns: Devnet only; no real funds; do not store sensitive personal data; use least-privilege permissions.

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